

Earthquake – Combined Testing From MST3 and MST4

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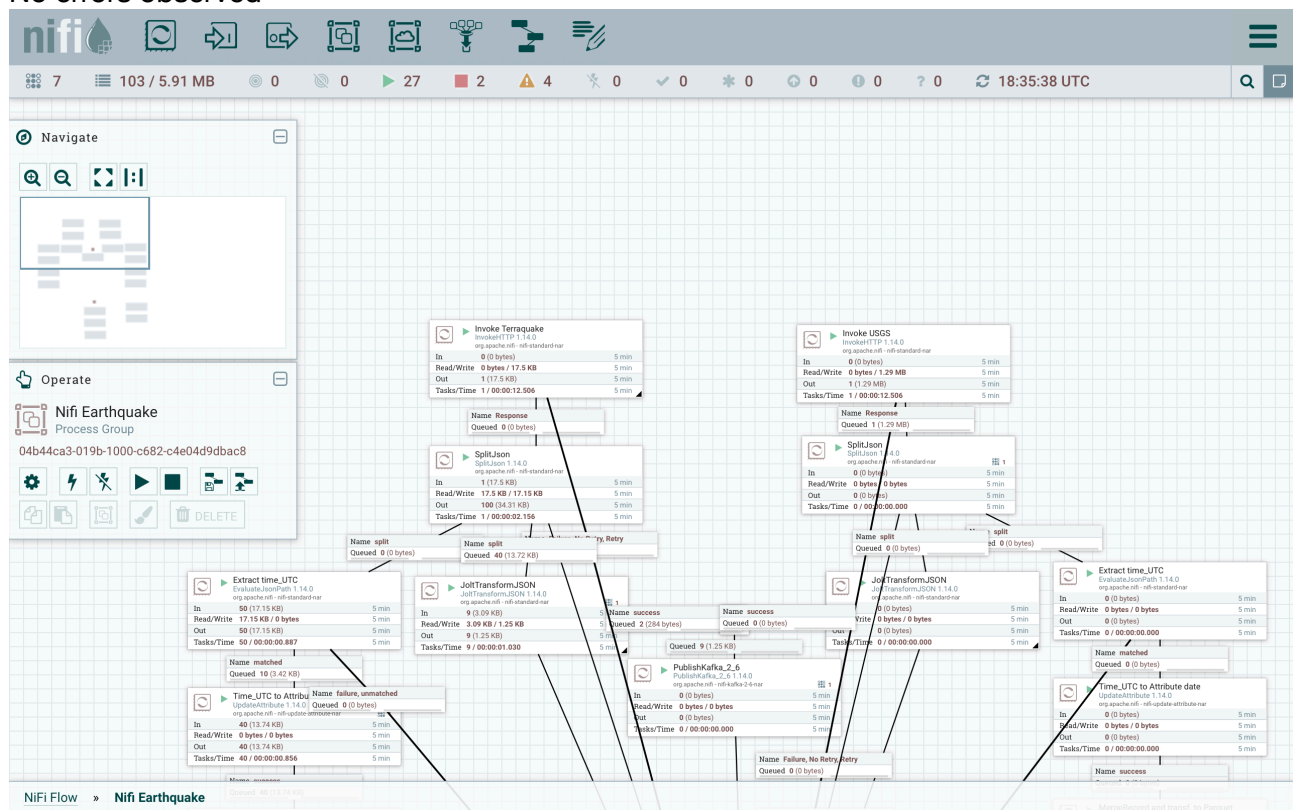
1. NiFi Flow Execution

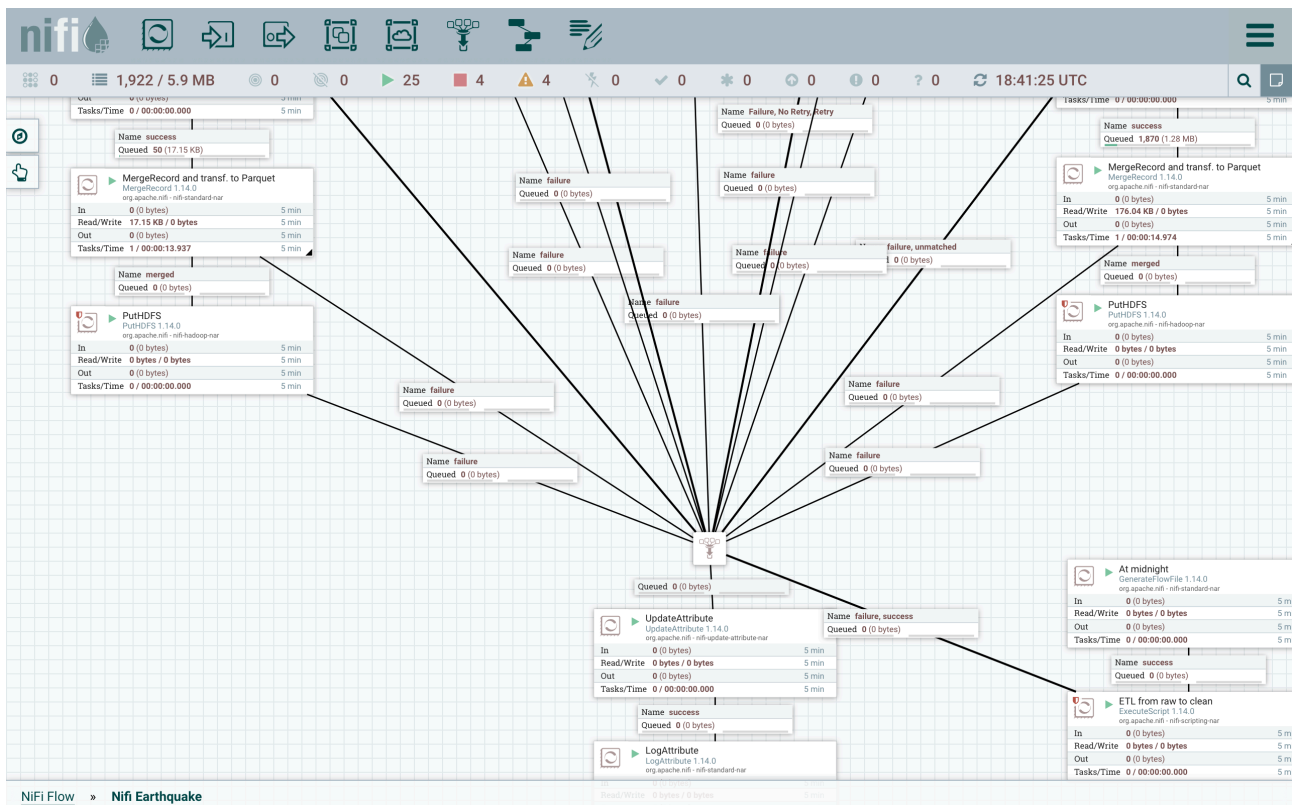
Objective: Verify that the NiFi flow executes successfully without errors.

Expected Result:

All processors are running

No errors observed





Result: PASS

2. Raw Data Ingestion

HDFS Path:

/user/testuser/earthquake_data/terraquake_raw

/user/testuser/earthquake_data/USGS_raw

```
[vagrant@node1:~]$ hdfs dfs -ls /user/testuser/earthquake_data/USGS_raw
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/usr/local/hadoop-2.7.6/share/hadoop/common/lib/slf4j-log4j12-1.7.10.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/usr/local/apache-tez-0.9.1-bin/lib/slf4j-log4j12-1.7.10.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.slf4j.impl.Log4jLoggerFactory]
Found 4 items
drwxr-xr-x - root supergroup          0 2025-12-17 18:24 /user/testuser/earthquake_data/USGS_raw/dt=2025-12-16
drwxr-xr-x - root supergroup          0 2025-12-17 18:24 /user/testuser/earthquake_data/USGS_raw/dt=2025-12-17
drwxr-xr-x - root supergroup          0 2025-12-22 18:45 /user/testuser/earthquake_data/USGS_raw/dt=2025-12-21
drwxr-xr-x - root supergroup          0 2025-12-22 18:45 /user/testuser/earthquake_data/USGS_raw/dt=2025-12-22
[vagrant@node1:~]$ hdfs dfs -ls /user/testuser/earthquake_data/terraquake_raw
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/usr/local/hadoop-2.7.6/share/hadoop/common/lib/slf4j-log4j12-1.7.10.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/usr/local/apache-tez-0.9.1-bin/lib/slf4j-log4j12-1.7.10.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.slf4j.impl.Log4jLoggerFactory]
Found 2 items
drwxr-xr-x - root supergroup          0 2025-12-17 18:18 /user/testuser/earthquake_data/terraquake_raw/dt=2025-12-17
drwxr-xr-x - root supergroup          0 2025-12-22 18:45 /user/testuser/earthquake_data/terraquake_raw/dt=2025-12-22
```

Objective: Validate that raw data is successfully ingested into HDFS.

Validation Checks:

Files exist in HDFS

Evidence (Screenshot):

Result: PASS

3. Kafka Event Registration

Objective: Verify that earthquake events are successfully registered in Kafka.

Validation Checks:

Kafka topic exists

Events are being produced/consumed

```
mikolajmalec — vagrant@node1: ~ — ssh -i ~/Big-data/LABOLATORIES/private_key vagrant@localhost -p 2222 — 167x24
vagrant@node1:~$ /usr/local/kafka/bin/kafka-console-consumer.sh --topic unified_quakes --bootstrap-server localhost:9092
{"id":"44835232","magnitude":1.3,"mag_type":"ML","place":"8 km NE Assisi (PG)","time_utc":"2025-12-22T13:47:49.590000","source":"terraqquake"}
{"id":"44835162","magnitude":1.6,"mag_type":"ML","place":"Costa Siciliana centro-settentrionale (Palermo)","time_utc":"2025-12-22T13:38:03.530000","source":"terraqquake"}
{"id":"44834542","magnitude":0.9,"mag_type":"ML","place":"1 km W Campotosto (AQ)","time_utc":"2025-12-22T11:53:12.290000","source":"terraqquake"}
{"id":"44834102","magnitude":6.4,"mag_type":"Mwp","place":"Papua New Guinea [Land]","time_utc":"2025-12-22T10:31:28.277000","source":"terraqquake"}
{"id":"44834002","magnitude":2.1,"mag_type":"ML","place":"3 km E Mangone (CS)","time_utc":"2025-12-22T10:23:20.800000","source":"terraqquake"}
{"id":"44833672","magnitude":1.1,"mag_type":"ML","place":"2 km NE Cascia (PG)","time_utc":"2025-12-22T09:21:01.750000","source":"terraqquake"}
{"id":"44833612","magnitude":0.5,"mag_type":"ML","place":"8 km N Locana (TO)","time_utc":"2025-12-22T08:57:27.230000","source":"terraqquake"}
{"id":"44833132","magnitude":1.3,"mag_type":"ML","place":"2 km S Ussita (MC)","time_utc":"2025-12-22T07:04:07.470000","source":"terraqquake"}
{"id":"44832862","magnitude":2.2,"mag_type":"ML","place":"Golfo di Salerno (Salerno)","time_utc":"2025-12-22T04:37:54.680000","source":"terraqquake"}
{"id":"44832832","magnitude":1.3,"mag_type":"ML","place":"6 km SE Cittareale (RI)","time_utc":"2025-12-22T03:25:06.520000","source":"terraqquake"}
{"id":"44832782","magnitude":1.2,"mag_type":"ML","place":"2 km NW Norcia (PG)","time_utc":"2025-12-22T02:45:57.760000","source":"terraqquake"}
{"id":"44832722","magnitude":1.2,"mag_type":"ML","place":"4 km E Cellara (CS)","time_utc":"2025-12-22T02:07:04.400000","source":"terraqquake"}
{"id":"44832682","magnitude":0.8,"mag_type":"ML","place":"4 km E Mangone (CS)","time_utc":"2025-12-22T01:21:09.270000","source":"terraqquake"}
{"id":"44832662","magnitude":0.4,"mag_type":"ML","place":"5 km NW Pietralunga (PG)","time_utc":"2025-12-22T01:05:09.400000","source":"terraqquake"}
{"id":"44832612","magnitude":1.2,"mag_type":"ML","place":"4 km E Mangone (CS)","time_utc":"2025-12-22T00:51:41.120000","source":"terraqquake"}
{"id":"44832572","magnitude":1.5,"mag_type":"ML","place":"2 km SW Mottafollone (CS)","time_utc":"2025-12-22T00:40:56.690000","source":"terraqquake"}
{"id":"44832522","magnitude":0.9,"mag_type":"ML","place":"5 km NE Cittareale (RI)","time_utc":"2025-12-22T00:40:26.610000","source":"terraqquake"}
{"id":"44832502","magnitude":1.2,"mag_type":"ML","place":"6 km E Gubbio (PG)","time_utc":"2025-12-21T23:23:57.410000","source":"terraqquake"}
{"id":"44832482","magnitude":1.2,"mag_type":"ML","place":"4 km E Cellara (CS)","time_utc":"2025-12-21T23:22:31.520000","source":"terraqquake"}
{"id":"44832432","magnitude":1.4,"mag_type":"ML","place":"7 km SE Bronte (CT)","time_utc":"2025-12-21T23:07:00.290000","source":"terraqquake"}
{"id":"44832382","magnitude":1.2,"mag_type":"ML","place":"7 km SE Bronte (CT)","time_utc":"2025-12-21T22:26:01.930000","source":"terraqquake"}
{"id":"44832312","magnitude":1.2,"mag_type":"ML","place":"4 km SE Pietralunga (PG)","time_utc":"2025-12-21T22:19:06.430000","source":"terraqquake"}
{"id":"44832282","magnitude":1,"mag_type":"ML","place":"3 km SE Campello sul Clitunno (PG)","time_utc":"2025-12-21T22:08:30.140000","source":"terraqquake"}
```

Result: PASS

4. Hive Raw Tables Validation

Objective: Validate Hive tables created on top of HDFS data.

Validation Checks:

Table column follow correct naming

Hive returns example data

```
mikolajmalec — vagrant@node1: ~ — ssh -i ~/Big-data/LABOLATORIES/private_key vagrant@localhost -p 2222 — 145x27
[hive> describe terraqquake_raw
. . > ;
OK
type string
properties struct<eventId:bigint,originId:bigint,time:string,author:string,magType:string,mag:double,magAuthor:string,type:string,place:string,version:int,geojson_creationTime:string>
geometry struct<type:string,coordinates:array<double>>
dt string

# Partition Information
# col_name          data_type          comment

dt string
9 rows selected (0.582 seconds)
[hive> describe usgs_raw;
OK
id string
properties struct<mag:double,place:string,time:bigint,title:string>
geometry struct<type:string,coordinates:array<double>>
dt string

# Partition Information
# col_name          data_type          comment

dt string
9 rows selected (0.696 seconds)
[hive> |

[hive> SELECT * FROM terraqquake_raw LIMIT 1;
OK
Feature {"eventId":44815002,"originId":141835741,"time":"2025-12-17T17:25:39.110000","author":"SURVEY-INGV","magType":"ML","mag":0.9,"magAuthor":"--","type":"earthquake","place":"4 km E Cagli (PU)","version":100,"geojson_creationTime":"2025-12-17T18:08:38"} {"type":"Point","coordinates":[12.6988,43.5598,18.0]} 2025-12-17
1 row selected (6.406 seconds)
[hive> SELECT * FROM usgs_raw LIMIT 1;
OK
uu80005863 {"mag":0.81,"place":"13 km N of Summit Park, Utah","time":1765929135360,"title":"M 0.8 - 13 km N of Summit Park, Utah"} {"type":"Point","coordinates":[-111.58233333333,40.868833333333,11.39]} 2025-12-16
1 row selected (1.629 seconds)
[hive> |
```

Result: PASS

5. Clean Data Availability

Objective: Verify that cleaned data is generated after the cleaning job.

Validation Checks:

Clean data exists in HDFS

```
vagrant@node1:~$ hdfs dfs -ls /user/testuser/earthquake_data/clean
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/usr/local/hadoop-2.7.6/share/hadoop/common/lib/slf4j-log4j12-1.7.10.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/usr/local/apache-tez-0.9.1-bin/lib/slf4j-log4j12-1.7.10.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.slf4j.impl.Log4jLoggerFactory]
Found 4 items
drwxr-xr-x - vagrant supergroup          0 2025-12-22 20:00 /user/testuser/earthquake_data/clean/dt=2025-12-16
drwxr-xr-x - vagrant supergroup          0 2025-12-22 20:00 /user/testuser/earthquake_data/clean/dt=2025-12-17
drwxr-xr-x - vagrant supergroup          0 2025-12-22 20:00 /user/testuser/earthquake_data/clean/dt=2025-12-21
drwxr-xr-x - vagrant supergroup          0 2025-12-22 20:00 /user/testuser/earthquake_data/clean/dt=2025-12-22
```

Result: PASS

6. Hive Clean Table Validation

Objective: Validate Hive tables created on top of HDFS data.

Validation Checks:

Table column follow correct naming

Queries return data

```
hive> DESCRIBE earthquake_clean;
OK
id string
magnitude double
mag_type string
place string
time_utc bigint
longitude double
latitude double
depth_km double
source_system string
dt string

# Partition Information
# col_name      data_type      comment
dt string
15 rows selected (0.449 seconds)
hive> SELECT * FROM earthquake_clean LIMIT 1;
OK
nc75280916 1.49 ml 9 km WNW of Templeton, CA 1765912519940 -120.800666809082 35.5833320617676 5.19999980926514 USGS 2025-12-16
1 row selected (2.381 seconds)
hive> |
```

Result: PASS

7. Historical Data Ingestion (USGS & Terraquake)

Objective: Verify that the Apache NiFi ingestion pipelines correctly retrieved the full historical dataset for the year 2025.

Test Steps:

1. Trigger the bulk ingestion process for the start date 2025-01-01 for USGS and Terraquake.
2. Inspect the HDFS Raw directories to confirm file presence for all months/days for both sources and for the clean data.
3. Using Hive, export example data from both sources.

Expected Result: A complete sequence of raw Parquet files covering the entire requested year without gaps and example hive output from the loaded tables.

Actual Result:

HDFS Raw directories output is stored in Test_appendix, they show the dates from 2025-01-01 to 2025-12-22 for both sources and for the clean data.

```
hive> SELECT * FROM usgs_raw LIMIT 1;
OK
tx2025abvp {"mag":1.6,"place":"35 km WSW of Ackerly, Texas","time":1735775714006,"title":"M 1.6 - 35 km WSW of Ackerly, Texas"} {"type":"Point","coordinates":[-102.059,32.401,8.374]} 2025-01-01
1 row selected (7.373 seconds)
hive> SELECT * FROM terraquake_raw LIMIT 1;
OK
Feature {"eventId":41269422,"originId":133577801,"time":"2025-01-01T22:47:10.750000","author":"SURVEY-INGV","magType":"ML","mag":1.5,"magAuthor":"--","type":"earthquake","place":"1 km E Robecco sul Naviglio (MI)","version":100,"geojson_creationTime":"2025-12-29T16:06:47"} {"type":"Point","coordinates":[8.9,45.4407,11.6]} 2025-01-01
1 row selected (2.429 seconds)
```

```
vagrant@node1:~$ cqlsh -e "SELECT id, place, region_id, cluster_id FROM earthquakes.events;"
```

id	place	region_id	cluster_id
ak2025znstnm	92 km SE of Kokhanok, Alaska	1	2
44853052	4 km NE Anghiari (AR)	9	93
us7000r16y	153 km S of Severo-Kuril'sk, Russia	5	50
44853082	2 km NE Cerreto di Spoleto (PG)	9	93
nc75287501	11 km ESE of Hidden Valley Lake, CA	2	6
44852952	5 km SW Frontone (PU)	9	93
nc75287496	4 km ESE of San Ramon, CA	2	142
us7000r16w	183 km WSW of San Vicente de Cañete, Peru	3	27
44852872	4 km S Lioni (AV)	9	22
44852822	4 km NE Anghiari (AR)	9	93
44852632	1 km E Santo Stefano di Magra (SP)	9	93
44852512	5 km SE Preci (PG)	9	93
44852152	5 km SW Frontone (PU)	9	93
44852092	4 km NE Anghiari (AR)	9	93
44851992	3 km NE Anghiari (AR)	9	93

(15 rows)

```
vagrant@node1:~$ cqlsh -e "SELECT id, place, region_id, cluster_id FROM earthquakes.events;"
```

id	place	region_id	cluster_id
44862232	Vesuvio	9	100
44862162	1 km NE Tavarnelle Val di Pesa (FI)	9	163
44862112	6 km W Sansepolcro (AR)	9	163
44862082	5 km W Sansepolcro (AR)	9	163
44862032	4 km NE Anghiari (AR)	9	163
44862002	5 km NE Anghiari (AR)	9	163
44861902	5 km NE Anghiari (AR)	9	163
44861872	5 km SW Preci (PG)	9	1
44861742	5 km W Sansepolcro (AR)	9	163
44861702	8 km E Città di Castello (PG)	9	1
ak2025znstnm	92 km SE of Kokhanok, Alaska	1	2
44853052	4 km NE Anghiari (AR)	9	93
us7000r16y	153 km S of Severo-Kuril'sk, Russia	5	50
44853082	2 km NE Cerreto di Spoleto (PG)	9	93
nc75287501	11 km ESE of Hidden Valley Lake, CA	2	6
44852952	5 km SW Frontone (PU)	9	93
nc75287496	4 km ESE of San Ramon, CA	2	142
us7000r16w	183 km WSW of San Vicente de Cañete, Peru	3	27
44852872	4 km S Lioni (AV)	9	22
44852822	4 km NE Anghiari (AR)	9	93
44852632	1 km E Santo Stefano di Magra (SP)	9	93
44852512	5 km SE Preci (PG)	9	93
44852152	5 km SW Frontone (PU)	9	93
44852092	4 km NE Anghiari (AR)	9	93
44851992	3 km NE Anghiari (AR)	9	93

(25 rows)

Status: Passed

8. Speed Layer Stream Enrichment

Objective: Confirm that real-time events processed by the Speed Layer are correctly enriched with geospatial metadata (Region ID and Cluster ID) and stored in Apache Cassandra.

Test Steps:

1. Observe the Kafka topic for a new incoming event.
2. Allow the Spark Streaming job to process the event.
3. Query the Apache Cassandra table earthquake_realtime to retrieve the record.

Expected Result: Kafka consumer will intercept the event in the JSON format and the record in Cassandra for the same event must have the region_id and cluster_id columns populated (not null).

Actual Result:

```
vagrant@node1:~$ /usr/local/kafka/bin/kafka-console-consumer.sh --topic unified_quakes --bootstrap-server localhost:9092
```

```
{
  "id": "44861872",
  "magnitude": 1.4,
  "mag_type": "ML",
  "place": "5 km SW Preci (PG)",
  "time_utc": "2025-12-29T21:46:41.240000",
  "longitude": 13.0058,
  "latitude": 42.8425,
  "depth_km": 7.7,
  "source": "terraquake"
}
```

```
{
  "id": "44861742",
  "magnitude": 0.9,
  "mag_type": "ML",
  "place": "5 km W Sansepolcro (AR)",
  "time_utc": "2025-12-29T20:54:12.270000",
  "longitude": 12.0875,
  "latitude": 43.5808,
  "depth_km": 9.8,
  "source": "terraquake"
}
```

```
{
  "id": "44861702",
  "magnitude": 1,
  "mag_type": "ML",
  "place": "8 km E Città di Castello (PG)",
  "time_utc": "2025-12-29T20:50:05.550000",
  "longitude": 12.3353,
  "latitude": 43.4812,
  "depth_km": 11.1,
  "source": "terraquake"
}
```

```
{
  "id": "44861682",
  "magnitude": 1.3,
  "mag_type": "ML",
  "place": "5 km W Sansepolcro (AR)",
  "time_utc": "2025-12-29T20:42:49.980000",
  "longitude": 12.0868,
  "latitude": 43.5843,
  "depth_km": 9.7,
  "source": "terraquake"
}
```

```
{
  "id": "44861632",
  "magnitude": 2.3,
  "mag_type": "ML",
  "place": "4 km NE Anghiari (AR)",
  "time_utc": "2025-12-29T20:41:43.740000",
  "longitude": 12.0887,
  "latitude": 43.5717,
  "depth_km": 9.3,
  "source": "terraquake"
}
```

```
{
  "id": "44861612",
  "magnitude": 0.6,
  "mag_type": "ML",
  "place": "5 km W Sansepolcro (AR)",
  "time_utc": "2025-12-29T20:34:04.600000",
  "longitude": 12.0807,
  "latitude": 43.5892,
  "depth_km": 10.1,
  "source": "terraquake"
}
```

```
{
  "id": "44861472",
  "magnitude": 1.2,
  "mag_type": "ML",
  "place": "6 km NE Claut (PN)",
  "time_utc": "2025-12-29T19:55:44.730000",
  "longitude": 12.5715,
  "latitude": 46.3835,
  "depth_km": 9.7,
  "source": "terraquake"
}
```

```
{
  "id": "44861292",
  "magnitude": 0.7,
  "mag_type": "ML",
  "place": "4 km NE Sellano (PG)",
  "time_utc": "2025-12-29T18:59:09.480000",
  "longitude": 12.9552,
  "latitude": 42.9173,
  "depth_km": 12.9,
  "source": "terraquake"
}
```

```
{
  "id": "44861172",
  "magnitude": 0.8,
  "mag_type": "ML",
  "place": "4 km NE Costacciaro (PG)",
  "time_utc": "2025-12-29T18:03:35.890000",
  "longitude": 12.7522,
  "latitude": 43.3762,
  "depth_km": 13,
  "source": "terraquake"
}
```

Status: Passed

9. Cross-Source Deduplication Validation

Objective: Verify that the Batch Layer deduplication logic successfully merged events from USGS and Terraquake. The Clean Layer should not contain any two events that are within 0.1 degrees of each other and within 10 minutes of each other.

Test Steps:

1. Execute the following QA query in Apache Hive to check for residual duplicates:

```
SELECT
-- IDs and Time
```

```

a.id AS event_1,
b.id AS event_2,
a.time_utc AS time_1,
b.time_utc AS time_2,

-- Sources
a.source_system AS source_1,
b.source_system AS source_2,

-- Magnitudes
a.magnitude AS mag_1,
b.magnitude AS mag_2,

-- Location & Diff
a.place,
(a.time_utc - b.time_utc) AS time_diff_ms

FROM earthquake_clean a
JOIN earthquake_clean b
ON a.dt = b.dt
WHERE
  -- 1. Prevent self-matches and duplicate pairs
  a.id < b.id

  -- 2. Ensure different sources
  AND a.source_system != b.source_system

  -- 3. Spatial filter (approx 0.1 degrees)
  AND ABS(a.latitude - b.latitude) <= 0.1
  AND ABS(a.longitude - b.longitude) <= 0.1

  -- 4. Time filter: 60,000 ms = 1 Minute
  AND ABS(a.time_utc - b.time_utc) <= 60000

LIMIT 10;

```

Expected Result: The query should return 0 rows, indicating that all overlapping events between the two sources were successfully merged into a single source of truth.

Actual Result:

```
hive> SELECT
-- IDs and Time
-- a.id AS event_1,
-- b.id AS event_2,
-- a.time_utc AS time_1,
-- b.time_utc AS time_2,
-- Sources
-- a.source_system AS source_1,
-- b.source_system AS source_2,
-- Magnitude
-- a.magnitude AS mag_1,
-- b.magnitude AS mag_2,
-- Location & Diff
-- a.place,
-- (a.time_utc - b.time_utc) AS time_diff_ms
FROM earthquake_clean a
JOIN earthquake_clean b
ON a.dt = b.dt
WHERE
-- 1. Prevent self-matches and duplicate pairs
a.id < b.id
-- 2. ENSURE DIFFERENT SOURCES (The new filter)
AND a.source_system != b.source_system
-- 3. Spatial filter (approx 0.1 degrees)
AND ABS(a.latitude - b.latitude) <= 0.1
AND ABS(a.longitude - b.longitude) <= 0.1
-- 4. Time filter: 60,000 ms = 1 Minute
AND ABS(a.time_utc - b.time_utc) <= 60000
LIMIT 10;
26/12/19 20:28:12 [HiveServer2-Background-Pool: Thread-176]: WARN ql.Driver: Hive-on-MR is deprecated in Hive 2 and may not be available in the future versions. Consider using a different execution engine (i.e. spark, tez) or using Hive 1.X releases.
WARNING: Hive-on-MR is deprecated in Hive 2 and may not be available in the future versions. Consider using a different execution engine (i.e. spark, tez) or using Hive 1.X releases.
Query ID = vagrant_202612192028080_f0fa8759-3282-4f2a-af38-65a358f8d81f
Total Jobs = 1
26/12/19 20:28:12 [HiveServer2-Background-Pool: Thread-176]: WARN ql.Driver: Hive-on-MR is deprecated in Hive 2 and may not be available in the future versions. Consider using a different execution engine (i.e. spark, tez) or using Hive 1.X releases.
WARNING: Hive-on-MR is deprecated in Hive 2 and may not be available in the future versions. Consider using a different execution engine (i.e. spark, tez) or using Hive 1.X releases.
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/usr/local/apache-hive-2.3.8-bin/lib/log4j-slf4j-impl-2.6.2.jar/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/usr/local/apache-tez-0.9.1-bin/lib/slf4j-log4j12-1.7.10.jar/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/usr/local/hadoop-2.7.0/share/hadoop/common/lib/slf4j-log4j12-1.7.10.jar/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.apache.logging.slf4j.Log4jLoggerFactory]
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/usr/local/apache-hive-2.3.8-bin/lib/log4j-slf4j-impl-2.6.2.jar/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/usr/local/apache-tez-0.9.1-bin/lib/slf4j-log4j12-1.7.10.jar/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/usr/local/hadoop-2.7.0/share/hadoop/common/lib/slf4j-log4j12-1.7.10.jar/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.apache.logging.slf4j.Log4jLoggerFactory]
2026-12-19 20:28:31 Starting to launch local task to process map join; maximum memory = 477626368
2026-12-19 20:28:31 Starting to launch local task to process map join; maximum memory = 477626368
SLF4J: Failed to load class 'org.slf4j.impl.StaticLoggerBinder'.
SLF4J: Defaulting to no-operation (NOP) logger implementation
SLF4J: See http://www.slf4j.org/codes.html#StaticLoggerBinder for further details.
SLF4J: Failed to load class 'org.slf4j.impl.StaticLoggerBinder'.
SLF4J: Defaulting to no-operation (NOP) logger implementation
SLF4J: See http://www.slf4j.org/codes.html#StaticLoggerBinder for further details.
2026-12-19 20:29:01 Dump the side-table for tag: 0 with group count: 363 into file: file:/tmp/vagrant/5c918366-5886-44c9-9883-46bf1204889d/hive_2025-12-19_20-28-00_458_2783438958487787782-1/-local-10084/HashTable-Stage-3/MapJoin-mapfile30---.hashtable
2026-12-19 20:29:02 Uploaded 1 file to: file:/tmp/vagrant/5c918366-5886-44c9-9883-46bf1204889d/hive_2025-12-19_20-28-00_458_2783438958487787782-1/-local-10084/HashTable-Stage-3/MapJoin-mapfile30---.hashtable (11473831 bytes)
2026-12-19 20:29:02 End of local task: Time taken: 31.710 sec.
Execution completed successfully
MapReduce task succeeded
Launching Job 3 out of 5
Number of reduce tasks is set to 0 since there's no reduce operator
2026-12-19 20:29:08 Dump the side-table for tag: 0 with group count: 363 into file: file:/tmp/vagrant/5c918366-5886-44c9-9883-46bf1204889d/hive_2025-12-19_20-28-00_458_2783438958487787782-1/-local-10084/HashTable-Stage-3/MapJoin-mapfile30---.hashtable
2026-12-19 20:29:02 Uploaded 1 file to: file:/tmp/vagrant/5c918366-5886-44c9-9883-46bf1204889d/hive_2025-12-19_20-28-00_458_2783438958487787782-1/-local-10084/HashTable-Stage-3/MapJoin-mapfile30---.hashtable (11473831 bytes)
2026-12-19 20:29:02 End of local task: Time taken: 31.716 sec.
26/12/19 20:29:04 [HiveServer2-Background-Pool: Thread-176]: WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
Starting Job = job_1767022533978_0015, Tracking URL = http://nodes10888/proxy/application/1767022533978_0015/
Kill Command = /usr/local/hadoop/bin/hadoop job -kill job_1767022533978_0015
Hadoop job information for Stage=1: number of maps: 0, number of reducers: 0
26/12/19 20:29:32 [HiveServer2-Background-Pool: Thread-176]: WARN mapreduce.Counters: Group org.apache.hadoop.mapred.Task$Counter is deprecated. Use org.apache.hadoop.mapreduce.Task$Counter instead
2026-12-19 20:29:32 295 Stage-3 map = 0N, reduce = 0N
26/12/19 20:29:32 [HiveServer2-Background-Pool: Thread-176]: WARN mapreduce.Counters: Group org.apache.hadoop.mapred.Task$Counter is deprecated. Use org.apache.hadoop.mapreduce.Task$Counter instead
2026-12-19 20:29:32 544 Stage-3 map = 0N, reduce = 0N
26/12/19 20:29:40 [HiveServer2-Background-Pool: Thread-176]: WARN mapreduce.Counters: Group org.apache.hadoop.mapred.Task$Counter is deprecated. Use org.apache.hadoop.mapreduce.Task$Counter instead
2026-12-19 20:29:40 245 Stage-3 map = 100N, reduce = 0N
26/12/19 20:29:40 [HiveServer2-Background-Pool: Thread-176]: WARN mapreduce.Counters: Group org.apache.hadoop.mapred.Task$Counter is deprecated. Use org.apache.hadoop.mapreduce.Task$Counter instead
End of Job = job_1767022533978_0015
MapReduce Jobs Launched:
26/12/19 20:29:40 [HiveServer2-Background-Pool: Thread-176]: WARN mapreduce.Counters: Group FileSystemCounter is deprecated. Use org.apache.hadoop.mapreduce.FileSystemCounter instead
Stage=Stage-3: HDFS Read: 0 HDFS Write: 0 SUCCESS
Total MapReduce CPU Time Spent: 0 msec
OK
No rows selected (169.21 seconds)
hive>
```

Total MapReduce CPU Time Spent: 0 msec

OK

No rows selected (169.21 seconds)

hive>

Status: Passed

10. K-Means Optimization & Stability Validation

Objective: Verify that the ML pipeline performs robust hyper-parameter tuning by iterating through a cluster range of $k = 300$ to 600 . The test ensures that for each k , 3 random-seed trials are executed to ensure stability, and the final model is automatically selected based on the highest Calinski-Harabasz (CH) Index score.

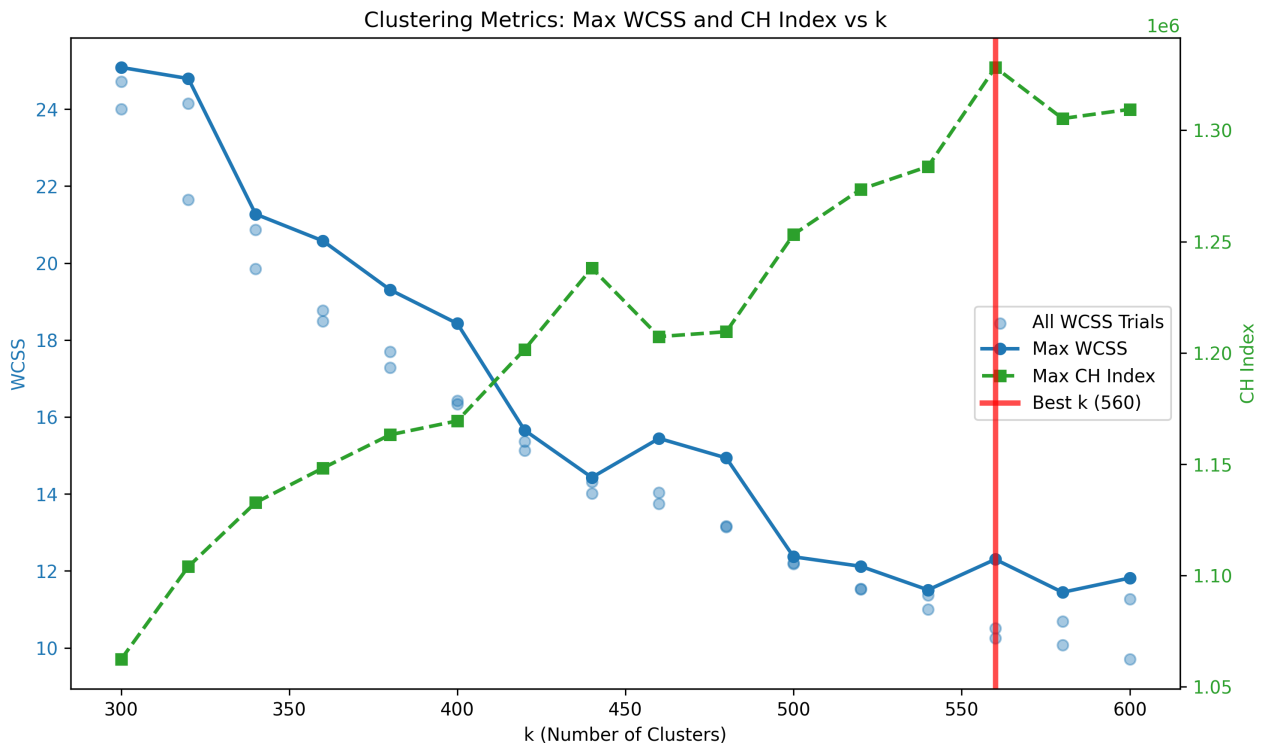
Test Steps:

- Review the PySpark training logs to confirm that multiple trials (iterations with different seeds) were executed for each value of k .
- Extract the performance metrics table listing k values (50–700) and their associated CH scores.
- Verify that the system selected the model corresponding to the maximum CH score.
- Plot the Within-Cluster Sum of Squares (WCSS) against the CH Index and observe the trend.

Expected Result:

- The logs must show multiple distinct WCSS values for every k , confirming that stability trials were run.
- The final selected k must correspond to the peak Calinski-Harabasz score.
- The selected k should geometrically align with the "elbow" (inflection point) of the WCSS curve, indicating where the variance reduction stabilises.

Actual Result:



Status: Passed

11. Centroid Back-Projection

Objective: Validate the mathematical accuracy of converting the machine learning model's 3D Cartesian centroids (X, Y, Z) back into Geographic coordinates (Latitude, Longitude) by performing a manual calculation check.

Test Steps:

1. Get a sample centroid vector from the model output.
2. Check if the model's output Lat/Lon matches the manual calculation.

Expected Result: The model output should match the manual calculation.

Actual Result:

```
hive> select * from dim_cluster_metadata limit 1;
OK
0 -0.2660453564903092 -0.6517040619291885 0.7095140480559485 45.22690837241916 -112.20677165688359 767.1438528108063
1 row selected (1.021 seconds)
```


$$r = \sqrt{x^2 + y^2 + z^2}$$

$$\phi = \arcsin\left(\frac{z}{r}\right)$$

$$x^2 \approx 0.07078$$

$$\phi = \arcsin\left(\frac{0.709514}{0.999454}\right) \approx \arcsin(0.70990)$$

$$y^2 \approx 0.42472$$

$$\phi \approx 0.7893 \text{ radians}$$

$$z^2 \approx 0.50341$$

Convert to degrees:

$$r = \sqrt{0.99891} \approx \mathbf{0.99945}$$

$$0.7893 \times \frac{180}{\pi} \approx \mathbf{45.223^\circ}$$

$$\lambda = \text{atan2}(-0.651704, -0.266045) \approx -1.9584 \text{ radians}$$

Convert to degrees:

$$-1.9584 \times \frac{180}{\pi} \approx \mathbf{-112.209^\circ}$$

The results are inside the rounding error $\pm 0.03^\circ$ which corresponds to ~ 300 meters

Status: Passed